

Chemistry, Derived

Class XII CBSE · Chemistry · Chapters 1–3

Chemistry, Derived

Twelve derivations, one algebraic move per line, each with the reason it is allowed. Every one ends in a formula that is already on Every Chemistry Formula — the point of this page is that you can rebuild them when memory fails, and that you can write the proof out when the paper asks for it.

- Derivations: 12
- Figures: 12
- Chapters: 3
- Marks covered: 42

How to use this

Chapters 4, 5 and 6 have no derivations worth the name — their marks come from recall and reasoning, and those live on the formula sheet. Everything derivable in this paper is on this page.

Read the **setup** first and draw the figure yourself before you look at the steps. A derivation you can only follow is not one you can write.

The *italic reason* beside a step is what turns it from an assertion into a proof. In the exam, those reasons are usually where the marks are.

CH 3 Chemical Kinetics — 5 derivations · 13 marks

D1 Integrated rate law for a zero order reaction — 2-3 marks

A reaction $R \longrightarrow P$ whose rate does not depend on the concentration of R at all. Start from the definition of rate, put the zero-order rate law into it, and integrate from the start of the reaction to time t . Write $[R]_0$ for the concentration at $t = 0$ and $[R]$ for the concentration at time t .

Figure. Graph of concentration of R against time for a zero order reaction: a straight line starting at the initial concentration on the vertical axis and falling with constant slope minus k , meeting the time axis when all of R is used up.

Zero order is the only order whose concentration-time plot is a straight line. Its slope is $-k$ and its intercept is $[R]_0$, which is how both are read off experimental data.

1. Rate $= -\frac{d[R]}{dt}$ — (definition; minus sign because $[R]$ falls)
2. Rate $= k[R]^0 = k$ — (zero order: anything to the power 0 is 1)
3. $-\frac{d[R]}{dt} = k$

4. $d[R] = -k dt$ — (separating the variables)
5. $\int_{[R]_0}^{[R]} d[R] = -k \int_0^t dt$ — (integrating between the two states)
6. $[R] - [R]_0 = -k(t - 0)$
7. $[R] - [R]_0 = -kt$
8. $[R] = -kt + [R]_0$ — (the straight line in the figure, $y = mx + c$)
9. $kt = [R]_0 - [R]$ — (rearranging for k)
10. $k = \frac{[R]_0 - [R]}{t}$

Result: $[R] = -kt + [R]_0 \quad k = \frac{[R]_0 - [R]}{t}$

Units check: k comes out as concentration $\div$ time, $\text{mol L}^{-1} \text{s}^{-1}$ — which is exactly what C3.4 predicts for $n = 0$.

D2 Half-life of a zero order reaction — 2 marks

Half-life $t_{1/2}$ is the time at which exactly half the reactant is left. Take the integrated law from D1 and substitute that one condition.

Shared setup with D1. Every half-life derivation is the integrated rate law with $[R] = [R]_0/2$ and $t = t_{1/2}$ substituted. If you can do D1 and D3, you can do D2 and D4 without learning anything new.

Figure. Zero order concentration against time, with the half-life marked: the concentration reaches half its initial value at a time equal to the initial concentration divided by twice k , and reaches zero at twice that time.

Because the line is straight, the reaction finishes at exactly $2t_{1/2}$ — a zero order reaction really does run out, which no first order reaction ever does.

1. $k = \frac{[R]_0 - [R]}{t}$ — (from D1)
2. At $t = t_{1/2}$, $[R] = \frac{[R]_0}{2}$ — (definition of half-life)
3. $k = \frac{[R]_0 - \frac{[R]_0}{2}}{t_{1/2}}$ — (substituting both)
4. $k = \frac{\frac{[R]_0}{2}}{t_{1/2}}$
5. $k t_{1/2} = \frac{[R]_0}{2}$
6. $t_{1/2} = \frac{[R]_0}{2k}$

Result: $t_{1/2} = \frac{[R]_0}{2k}$ — directly proportional to the initial concentration

D3 Integrated rate law for a first order reaction — 3 marks

Same reaction $R \rightarrow P$, but now the rate is proportional to $[R]$. The integration produces a natural logarithm, which is then converted to base 10 because every exam table and graph uses $\log$.

Figure. Graph of log of concentration of R against time for a first order reaction: a straight line with intercept log of the initial concentration and slope minus k over 2.303.

A first order reaction is straight only after you take the logarithm. That is the practical test: if $\log[R]$ against t is a line, the reaction is first order and k is $2.303 \times$ the magnitude of its slope.

1. $\text{Rate} = -\frac{d[R]}{dt}$ — (definition)
2. $\text{Rate} = k[R]^1 = k[R]$ — (first order rate law)
3. $-\frac{d[R]}{dt} = k[R]$
4. $\frac{d[R]}{[R]} = -k dt$ — (separating the variables)
5. $\int_{[R]_0}^{[R]} \frac{d[R]}{[R]} = -k \int_0^t dt$
6. $\ln[R] - \ln[R]_0 = -kt$ — (integral of $1/x$ is $\ln x$)
7. $\ln \frac{[R]}{[R]_0} = -kt$ — (law of logarithms)
8. $\ln \frac{[R]_0}{[R]} = kt$ — (inverting the fraction flips the sign)
9. $2.303 \log \frac{[R]_0}{[R]} = kt$ — ($\ln x = 2.303 \log_{10} x$)
10. $k = \frac{2.303}{t} \log \frac{[R]_0}{[R]}$
11. Rearranged for a graph: $\log[R] = -\frac{k}{2.303}t + \log[R]_0$ — (the straight line drawn above)

Result: $k = \frac{2.303}{t} \log \frac{[R]_0}{[R]}$

Units check: the log is dimensionless, so k has units of $1/\text{time}$, s^{-1} — matching C3.4 for $n = 1$. That is also why the next derivation comes out independent of concentration.

D4 First order half-life, and why it ignores concentration — 2-3 marks

Substitute the half-life condition into D3. Watch what happens to $[R]_0$ — the whole point of this derivation is that it cancels.

Figure. First order decay curve: concentration falls from its initial value to one half after one half-life, to one quarter after two half-lives and one eighth after three, with the three time intervals all equal.

Each successive halving takes the same time, no matter how little is left. That is the visual statement of the algebraic cancellation below, and it is why radioactive decay has a fixed half-life.

1. $k = \frac{2.303}{t} \log \frac{[R]_0}{[R]}$ — (from D3)
2. At $t = t_{1/2}$, $[R] = \frac{[R]_0}{2}$ — (definition of half-life)
3. $k = \frac{2.303}{t_{1/2}} \log \frac{[R]_0}{[R]_0/2}$
4. $\frac{[R]_0}{[R]_0/2} = 2$ — ($[R]_0$ cancels — this is the whole result)

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$$5. k = \frac{2.303}{t_{1/2}} \log 2$$

$$6. \log 2 = 0.301$$

$$7. k = \frac{2.303 \times 0.301}{t_{1/2}} = \frac{0.693}{t_{1/2}}$$

$$8. t_{1/2} = \frac{0.693}{k}$$

Result: $t_{1/2} = \frac{0.693}{k}$ — independent of the initial concentration

Step 4 is the answer to "show that the half-life of a first order reaction is independent of initial concentration". Do not skip it; it is the mark.

D5 Activation energy from rate constants at two temperatures — 3 marks

Start from the Arrhenius equation, take logarithms to make it linear, write it once at each of two temperatures T_1 and T_2 , and subtract. The frequency factor A is unknown and must disappear — that is what the subtraction is for.

Figure. Arrhenius plot: $\log k$ against one over T is a descending straight line whose intercept is $\log A$ and whose slope is minus the activation energy divided by $2.303 R$.

$T_2 > T_1$, so $1/T_2$ sits to the left of $1/T_1$ and the higher temperature gives the higher $\log k$. Reading the axis backwards is the commonest error on this plot.

$$1. k = A e^{-E_a/RT} \text{ — (Arrhenius equation)}$$

$$2. \ln k = \ln A - \frac{E_a}{RT} \text{ — (taking natural logs of both sides)}$$

$$3. \log k = \log A - \frac{E_a}{2.303RT} \text{ — (converting to base 10)}$$

$$4. \text{At } T_1: \log k_1 = \log A - \frac{E_a}{2.303RT_1}$$

$$5. \text{At } T_2: \log k_2 = \log A - \frac{E_a}{2.303RT_2}$$

$$6. \text{Subtracting: } \log k_2 - \log k_1 = -\frac{E_a}{2.303RT_2} + \frac{E_a}{2.303RT_1} \text{ — (log } A \text{ cancels — it is the same constant at both temperatures)}$$

$$7. \log \frac{k_2}{k_1} = \frac{E_a}{2.303R} \left[\frac{1}{T_1} - \frac{1}{T_2} \right]$$

$$8. \frac{1}{T_1} - \frac{1}{T_2} = \frac{T_2 - T_1}{T_1 T_2} \text{ — (common denominator)}$$

$$9. \log \frac{k_2}{k_1} = \frac{E_a}{2.303R} \left[\frac{T_2 - T_1}{T_1 T_2} \right]$$

Result: $\log \frac{k_2}{k_1} = \frac{E_a}{2.303R} \left[\frac{T_2 - T_1}{T_1 T_2} \right]$

The bracket is $\frac{1}{T_1} - \frac{1}{T_2}$, smaller temperature first. If you get a negative E_a , you reversed it at step 7.

CH 2 Electrochemistry — 3 derivations · 14 marks

D6 The Nernst equation — 3 marks

A galvanic cell running the reaction $aA + bB \rightarrow cC + dD$ with n electrons transferred. Two facts are combined: the thermodynamic relation between Gibbs energy and the reaction quotient, and the electrochemical relation between Gibbs energy and cell potential.

Figure. A galvanic cell: a zinc electrode in zinc sulphate solution on the left as the anode where oxidation occurs, a copper electrode in copper sulphate solution on the right as the cathode where reduction occurs, joined by a salt bridge that dips into both solutions and by an external circuit with a voltmeter through which electrons flow from left to right.

The Daniell cell, $E_{\text{cell}}^{\circ} = 1.1 \text{ V}$. The Nernst equation says what happens to that reading when the two solutions are not at 1 M — and the sign of the correction follows from Le Chatelier, not from memory.

1. $\Delta_r G = \Delta_r G^{\circ} + RT \ln Q$ — (thermodynamics: Gibbs energy away from standard state)

2. $\Delta_r G = -nFE_{\text{cell}}$ — (electrical work done by the cell)

3. $\Delta_r G^{\circ} = -nFE_{\text{cell}}^{\circ}$ — (same relation at standard state)

4. $-nFE_{\text{cell}} = -nFE_{\text{cell}}^{\circ} + RT \ln Q$ — (substituting 2 and 3 into 1)

5. Divide throughout by $-nF$:

$$6. E_{\text{cell}} = E_{\text{cell}}^{\circ} - \frac{RT}{nF} \ln Q$$

$$7. E_{\text{cell}} = E_{\text{cell}}^{\circ} - \frac{2.303RT}{nF} \log Q \quad \text{— (converting to base 10)}$$

$$8. \text{At } 298 \text{ K: } \frac{2.303 \times 8.314 \times 298}{96487} = 0.059 \quad \text{— (substituting } R, T \text{ and } F)$$

$$9. E_{\text{cell}} = E_{\text{cell}}^{\circ} - \frac{0.059}{n} \log Q$$

Result: $E_{\text{cell}} = E_{\text{cell}}^{\circ} - \frac{2.303RT}{nF} \log Q \xrightarrow{298 \text{ K}} E_{\text{cell}}^{\circ} - \frac{0.059}{n} \log Q$

Step 8 is worth writing out in full — being asked where the 0.059 comes from is a common follow-up, and "2.303RT/F at 298 K" is the whole answer.

D7 Equilibrium constant from standard cell potential — 2-3 marks

Take the Nernst equation and ask what it says at the one moment the cell stops working. A cell at equilibrium has no driving force, so it reads zero volts — and at that moment the reaction quotient has become the equilibrium constant.

Shared setup with D6. This is the Nernst equation with one substitution. Do not learn it as a separate formula.

Figure. Graph of cell potential against $\log Q$: a descending straight line of slope minus 0.059 over n , with intercept the standard cell potential at $\log Q$ equals zero, crossing the horizontal axis at $\log$ of the equilibrium constant where the cell is dead.

The cell potential falls linearly as products build up. Where the line crosses zero, Q has reached K_c — the battery is flat. Reading the intercept and the zero-crossing off this one line gives both results.

1. $E_{\text{cell}} = E_{\text{cell}}^{\circ} - \frac{0.059}{n} \log Q$ — (from D6)
2. At equilibrium the cell delivers no work, so $E_{\text{cell}} = 0$ — (a dead cell)
3. At equilibrium $Q = K_c$ — (definition of the equilibrium constant)
4. $0 = E_{\text{cell}}^{\circ} - \frac{0.059}{n} \log K_c$
5. $E_{\text{cell}}^{\circ} = \frac{0.059}{n} \log K_c$ — (moving the log term across — this is why the sign is now positive)
6. $\log K_c = \frac{n E_{\text{cell}}^{\circ}}{0.059}$ — (rearranged for K_c)
7. Equivalently, from $\Delta_r G^{\circ} = -nFE_{\text{cell}}^{\circ}$ and $\Delta_r G^{\circ} = -RT \ln K_c$:
8. $nFE_{\text{cell}}^{\circ} = RT \ln K_c$ — (equating the two expressions for $\Delta_r G^{\circ}$)

Result: $E_{\text{cell}}^{\circ} = \frac{0.059}{n} \log K_c$ $\log K_c = \frac{n E_{\text{cell}}^{\circ}}{0.059}$

D8 Molar conductivity from conductivity — where the 1000 comes from — 2 marks

A conductivity cell of electrode area A and separation l . Conductivity κ is defined for a 1 cm cube of solution; molar conductivity Λ_m is defined for however much solution contains exactly one mole. The 1000 is nothing but a cm^3 -to-litre conversion, and this derivation is really a units argument.

Figure. A conductivity cell: two parallel electrodes of area A separated by a distance l , with the cell constant defined as l divided by A , beside a unit cube of solution of side one centimetre whose conductance is the conductivity.

Two different reference volumes: κ is per cubic centimetre, Λ_m is per mole. Converting between them is the entire content of the formula.

1. $\Lambda_m = \kappa \times V$ — (V = volume of solution containing exactly 1 mole of electrolyte)
2. Let the concentration be c mol per litre — (the unit every exam question uses)
3. c moles occupy 1 L, so 1 mole occupies $\frac{1}{c}$ L
4. 1 L = 1000 cm^3 — (the conversion — this is the 1000)
5. 1 mole occupies $\frac{1000}{c} \text{cm}^3$
6. $\Lambda_m = \kappa \times \frac{1000}{c}$
7. Units: $\text{S cm}^{-1} \times \text{cm}^3 \text{mol}^{-1} = \text{S cm}^2 \text{mol}^{-1}$ — (confirming the standard unit)

Result: $\Lambda_m = \frac{\kappa \times 1000}{c}$ in $\text{S cm}^2 \text{mol}^{-1}$

In strict SI (κ in S m^{-1} , c in mol m^{-3}) step 4 disappears and $\Lambda_m = \kappa/c$ in $\text{S m}^2 \text{mol}^{-1}$. The 1000 is a unit artefact, not physics.

CH 1 Solutions — 4 derivations · 15 marks

D9 Relative lowering of vapour pressure equals the solute's mole fraction — 3 marks

A non-volatile solute B dissolved in a volatile solvent A. Only A can evaporate, so the vapour above the solution is pure A — but there is less of it, because solute particles occupy part of the surface. Start from Raoult's law and rearrange.

Figure. Two beakers side by side: pure solvent on the left with many molecules escaping from its surface, and a solution on the right whose surface is partly occupied by solute particles so fewer solvent molecules escape, giving a lower vapour pressure.

Solute particles (filled circles) take up surface sites, so fewer solvent molecules can leave. The fraction of the surface still available to solvent is x_A — which is exactly what Raoult's law asserts.

1. $p_A = x_A p_A^\circ$ — (Raoult's law; the solute is non-volatile so $p_{\text{soln}} = p_A$)
2. $p_A^\circ - p_A = p_A^\circ - x_A p_A^\circ$ — (subtracting both sides from p_A°)
3. $p_A^\circ - p_A = p_A^\circ(1 - x_A)$
4. $x_A + x_B = 1$, so $1 - x_A = x_B$ — (mole fractions sum to one)
5. $p_A^\circ - p_A = p_A^\circ x_B$
6. $\frac{p_A^\circ - p_A}{p_A^\circ} = x_B$ — (dividing by p_A° — the left side is the relative lowering)
7. $x_B = \frac{n_B}{n_A + n_B}$ — (definition of mole fraction)
8. For a dilute solution $n_B \ll n_A$, so $n_A + n_B \approx n_A$
9. $\frac{p_A^\circ - p_A}{p_A^\circ} \approx \frac{n_B}{n_A} = \frac{w_B/M_B}{w_A/M_A}$
10. $\frac{p_A^\circ - p_A}{p_A^\circ} = \frac{w_B M_A}{M_B w_A}$ — (rearranged to give molar mass)

Result: $\frac{p_A^\circ - p_A}{p_A^\circ} = x_B \quad M_B = \frac{w_B M_A p_A^\circ}{w_A (p_A^\circ - p_A)}$

Steps 1–6 are exact. Step 8 is the only approximation, and it is only valid for a dilute solution — say so if you use it.

D10 Elevation of boiling point is proportional to molality — 3 marks

A liquid boils when its vapour pressure equals the external pressure. A non-volatile solute lowers the vapour pressure at every temperature (D9), so the solution's curve sits below the solvent's — and has to be taken to a higher temperature before it reaches the 1 atm line.

Figure. Vapour pressure against temperature for pure solvent and for solution: the solution curve lies below the solvent curve everywhere, so it meets the one atmosphere line at a higher temperature, and the gap between the two crossing points is the elevation of boiling point.

The dashed curve is the solution. It never reaches 1 atm at the solvent's boiling point, so the temperature must be raised by ΔT_b — and for a dilute solution the two curves are near enough parallel that the shift is proportional to how much vapour pressure was lost.

1. For a dilute solution, the lowering of vapour pressure is proportional to the elevation of boiling point: $\Delta T_b \propto (p_A^\circ - p_A)$ — (the two curves are almost straight and almost parallel over this small range)
2. $\frac{p_A^\circ - p_A}{p_A^\circ} = x_B$ — (from D9)
3. So $\Delta T_b \propto x_B$ — (p_A° is a constant for a given solvent)
4. $x_B = \frac{n_B}{n_A + n_B} \approx \frac{n_B}{n_A}$ — (dilute solution)
5. $\Delta T_b \propto \frac{n_B}{n_A}$
6. $n_A = \frac{w_A}{M_A}$, so $\frac{n_B}{n_A} = \frac{n_B M_A}{w_A}$
7. M_A is a constant for a given solvent, so $\Delta T_b \propto \frac{n_B}{w_A}$
8. $\frac{n_B}{w_A \text{ in kg}} = m$, the molality — (definition of molality)
9. $\Delta T_b \propto m$
10. $\Delta T_b = K_b m$ — (introducing the proportionality constant)
11. $m = \frac{w_B \times 1000}{M_B w_A}$ — (molality written from masses, w_A in grams)
12. $\Delta T_b = \frac{K_b w_B \times 1000}{M_B w_A}$
13. $M_B = \frac{K_b w_B \times 1000}{\Delta T_b w_A}$

Result: $\Delta T_b = K_b m$ $M_B = \frac{K_b w_B \times 1000}{\Delta T_b w_A}$

K_b is the elevation for a 1 molal solution, and depends only on the solvent — for water $0.52 \text{ K kg mol}^{-1}$.

D11 Depression of freezing point is proportional to molality — 3 marks

A solution freezes when the vapour pressure of the liquid equals that of the pure solid solvent. Adding solute lowers the liquid's vapour pressure, so the two curves now meet at a lower temperature. Structurally identical to D10 — only the meeting point changes.

Shared setup with D10. Both start from "vapour pressure is lowered by x_B " and end in "proportional to molality". If you can write one, change the meeting condition and you have written the other.

Figure. Vapour pressure against temperature near the freezing point: the solid solvent line crosses the pure solvent curve at the normal freezing point on the right, and crosses the lower solution curve at a lower temperature on the left, the gap between the two crossings being the depression of freezing point.

Freezing happens where the liquid curve crosses the solid curve. Lowering the liquid curve moves that crossing to the left, and the shift is ΔT_f . Note the solid curve is unchanged — solute does not dissolve in the solid solvent.

1. A solution freezes when the vapour pressure of the liquid equals that of the pure solid solvent — (definition of freezing point)
2. For a dilute solution, $\Delta T_f \propto (p_A^\circ - p_A)$ — (same near-linear argument as D10)

3. $\frac{p_A^\circ - p_A}{p_A^\circ} = x_B$ — (from D9)
4. $\Delta T_f \propto x_B$
5. $x_B \approx \frac{n_B}{n_A}$ — (dilute solution)
6. $\Delta T_f \propto \frac{n_B}{w_A}$ — (M_A constant for a given solvent)
7. $\Delta T_f \propto m$ — (definition of molality)
8. $\Delta T_f = K_f m$
9. $m = \frac{w_B \times 1000}{M_B w_A}$
10. $M_B = \frac{K_f w_B \times 1000}{\Delta T_f w_A}$

Result: $\Delta T_f = K_f m$ $M_B = \frac{K_f w_B \times 1000}{\Delta T_f w_A}$

K_f for water is $1.86 \text{ K kg mol}^{-1}$ — larger than K_b , which is why freezing-point depression is the more sensitive method for finding a molar mass.

D12 Degree of dissociation from the van't Hoff factor — 2-3 marks

One mole of an electrolyte A_n that breaks into n particles, with a fraction α actually dissociating. Colligative properties count *particles*, so build the particle total after dissociation and compare it with the total before.

Figure. Particle count before and after dissociation: one mole of solute starts as one particle, and after a fraction alpha dissociates into n particles the total becomes one minus alpha plus n alpha, whose ratio to the original is the van't Hoff factor.

Only the fraction α splits; the rest stays whole. Counting both contributions is the entire derivation — i is just the ratio of the two particle totals.

1. $A_n \longrightarrow n A$, starting from 1 mole with a fraction α dissociating
2. Moles of A_n left undissociated = $1 - \alpha$
3. Moles of A produced = $n\alpha$ — (each dissociated mole gives n particles)
4. Total moles of particles after = $(1 - \alpha) + n\alpha$
5. $i = \frac{\text{total particles after}}{\text{total particles before}} = \frac{1 - \alpha + n\alpha}{1}$ — (definition of the van't Hoff factor)
6. $i = 1 - \alpha + n\alpha$
7. $i - 1 = \alpha(n - 1)$ — (collecting the α terms)
8. $\alpha = \frac{i - 1}{n - 1}$
9. For **association**, n molecules combine into 1, so the total after is $1 - \alpha + \frac{\alpha}{n}$
10. $i = 1 - \alpha + \frac{\alpha}{n} = 1 - \alpha \left(1 - \frac{1}{n}\right)$
11. $\alpha = \frac{1 - i}{1 - \frac{1}{n}}$

Result: $\alpha_{\text{dissoc}} = \frac{i - 1}{n - 1}$ $\alpha_{\text{assoc}} = \frac{1 - i}{1 - \frac{1}{n}}$

Dissociation gives $i > 1$ and association $i < 1$; both formulas return a positive α because the numerator flips with them.

Built from Sourabh Raina's one-shot and PYQ videos for Chapters 1-3, and cross-checked against NCERT Class XII Chemistry (Rationalised 2022-23) for constants and conventions — $F = 96487 \text{ C mol}_1^-$, $R = 8.314 \text{ J K}_1^- \text{ mol}_1^-$, $K_b = 0.52$ and $K_f = 1.86 \text{ K kg mol}_1^-$ for water, and **0.059** rather than 0.0591 in the Nernst equation.

Chapters 4, 5 and 6 are not represented here because they contain no derivations — their marks come from trends, reasons and reactivity orders, all of which are on the companion page, **Every Chemistry Formula**.